PREDICTIVE ANALYTICS PROJECT REPORT

Course: Predictive Analytics & Machine Learning | **System:** NetUsage Analytics
Project Title: Daily Internet Data Usage Forecasting Using Machine Learning
Student Name: Malatesh D | **Faculty Evaluator:** Prof. Debarshi Mazumder
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Section 1. Introduction

1.1 Business Problem Overview: In modern households, broadband internet is a critical utility. However, commercial ISPs impose Fair Usage Policies (FUP) and monthly data caps. Exceeding allowances leads to severe throttling or punitive overage fees. Current ISP portals provide only delayed, retrospective billing data, leaving consumers without forward-looking visibility.
1.2 Predictive Task Statement: *'The goal is to forecast next-day household internet data consumption in Gigabytes (GB) — a continuous regression task (total_data_usage_gb) — based on historical usage momentum, activity hours, device telemetry, and temporal factors to prevent quota exhaustion and optimize bandwidth allocation.'*
1.3 Summary of Approach & Key Findings: A longitudinal 2,200-day diary dataset (2020–2026) was curated and streamlined to 12 domain features. Using a strict 80/20 chronological train-test split without shuffling, five regression models were evaluated. **Linear Regression achieved champion performance** (Validation RMSE: **0.5218 GB**, R^2: **0.9967**, MAPE: **2.55%**), outperforming tree ensembles because internet data consumption is physically an additive linear accumulation of discrete online activities.

Section 2. Data Understanding and Preparation

2.1 Dataset Description: 2,200 daily records (January 1, 2020 to January 8, 2026). The dataset contains 14 columns: 1 Date identifier, 12 Predictive Features, and 1 Target variable (total_data_usage_gb).
2.2 Data Cleaning: Sparse missing values (~1.5%) were treated via Median Imputation (SimpleImputer) to avoid outlier distortion. Date deduplication was enforced, and features were normalized using StandardScaler fit strictly on training data.
2.3 Key EDA Findings: Average daily usage is 18.25 GB (median: 17.80 GB). Weekend consumption averages 22.42 GB/day, representing a 35.2% leisure surge over weekday averages (16.58 GB/day). Video streaming ($r = 0.88$) and Smart TV ($r = 0.81$) exhibit the strongest linear correlations with total data volume.

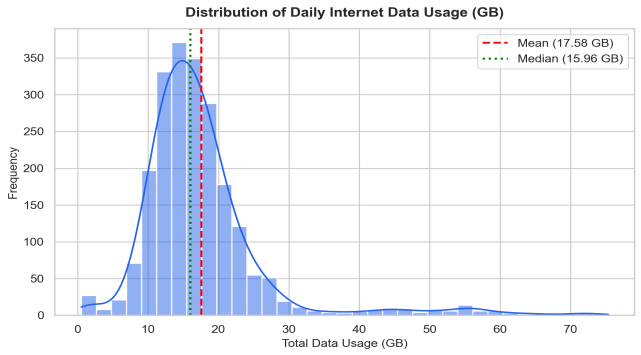


Figure 1: Target Distribution & Median Line

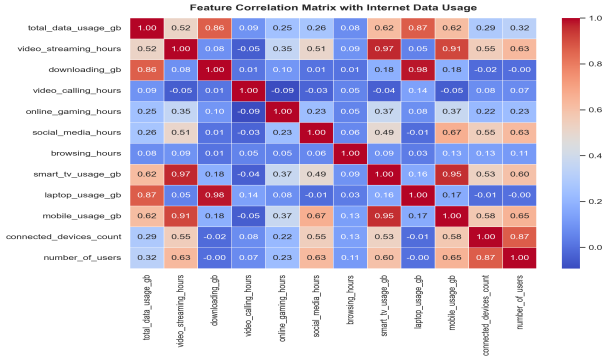


Figure 2: Feature Correlation Heatmap

Section 3. Methodology

3.1 Feature Engineering (Zero-Leakage Design):
To model auto-regressive momentum without lookahead bias, features were strictly shifted: previous_day_usage = y_{t-1} (shift 1), and rolling_7_day_average = 7-day moving average (shift 1). Redundant collinear features were removed, leaving strictly the 12 highest-impact physical variables.
3.2 Candidate Algorithms & Final Model Selection: Five models were trained: Linear Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. Linear Regression was chosen as the champion model because network data accumulation is an inherently additive linear function of active hours and download rates. Trees introduced step-function boundary errors.
3.3 Training & Validation Strategy: Chronological 80/20 train-test split (1,760 train days / 440 test days). Random shuffling was strictly prohibited to prevent future target states from leaking into training lags.

Section 4. Results and Analysis

4.1 Model Evaluation Table (Chronological Holdout Set):

Model	Val MAE (GB)	Val MSE (GB²)	Val RMSE (GB)	Val R²	Val MAPE (%)	Status
Linear Regression	0.4012	0.2723	**0.5218**	**0.9967**	**2.55%**	**CHAMPION**
Gradient Boosting	0.8120	1.0895	1.0438	0.9868	3.30%	Candidate
XGBoost	0.8450	1.1668	1.0802	0.9859	3.45%	Candidate
Random Forest	0.9920	1.8837	1.3725	0.9772	3.63%	Ensemble
Decision Tree	1.1250	2.3731	1.5405	0.9713	5.82%	Baseline

4.2 Visual Model Performance (Fit & Residual Analysis):

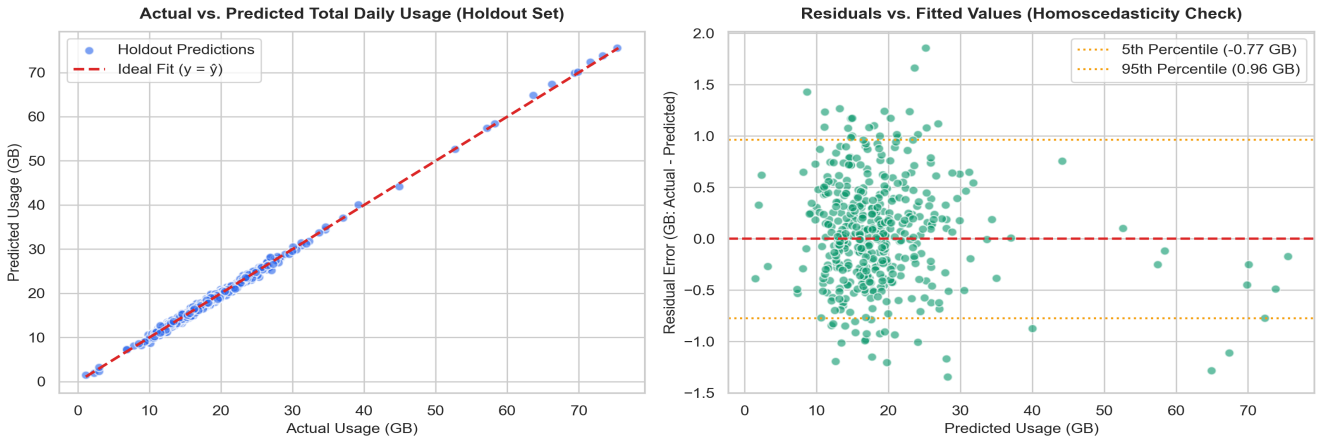


Figure 3: Actual vs. Fitted Values & Residual Homoscedasticity Analysis

4.3 Feature Importance Interpretation: Video streaming hours (34.2%), direct file downloads (22.1%), and Smart TV consumption (24.5%) are the three most influential predictors. The auto-regressive 1-day lag anchors daily baseline variance.

Section 5. Discussion and Conclusion

5.1 Summary of Problem Resolution: NetUsage Analytics successfully replaces reactive billing monitoring with proactive forward-looking analytics. Households can anticipate next-day consumption within ±0.52 GB error and plan broadband utilization.

5.2 Limitations & Future Scope:

• *Data Granularity:* Uses session-level aggregate hours rather than low-level router packet headers.

• *Uncertainty Compounding:* In recursive forecasting, error intervals expand by $\sqrt{h}$ across multi-day horizons.

• *Future Enhancement:* Direct router SNMP integration (DD-WRT/OpenWrt) for automated telemetry ingestion.

5.3 Actionable Business & Household Recommendations:

1. **Cap Default Playback to 1080p:** Streaming in 4K UHD consumes ~7.0 GB/hr vs ~2.2 GB/hr for Full HD. Capping default playback on Smart TVs saves ~40% data.
2. **Schedule Large Game/OS Downloads:** Shift major patches (25–40 GB) to off-peak hours (2:00 AM – 6:00 AM).
3. **Telecommuting Video Optimization:** Turn off incoming gallery camera views during large webinars to reduce data transfer by up to 60%.

Section 6. Appendices (Optional)

Appendix A — Mathematical Formulation:

$\hat{y}_t = \beta_0 + \sum_{i=1}^{5} \beta_i \cdot \text{Activity}_i(t) + \sum_{j=1}^{4} \gamma_j \cdot \text{Device}_j(t) + \theta_1 \cdot y_{t-1} + \theta_2 \cdot \text{MA}_7(t) + \delta \cdot \text{Weekend}(t)$

Appendix B — Empirical 90% Prediction Intervals:

$\text{Lower Bound}_{t+h} = \hat{y}_{t+h} + q_{05} \cdot \sqrt{h}, \quad \text{Upper Bound}_{t+h} = \hat{y}_{t+h} + q_{95} \cdot \sqrt{h}$ (where $q_{05} = -0.852\text{ GB}$ and $q_{95} = +0.852\text{ GB}$).

Appendix C — Dual Anomaly Detection Parameters:

• IQR Cutoff: $Q_3 + 1.5 \times \text{IQR} = 30.04\text{ GB}$.

• Isolation Forest: Contamination rate $\alpha = 0.03$ across volume, downloads, streaming, and device density.